COMP 1111 2. Midterm

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I. QUESTION (16 POINTS)

When you use an ATM with your bank card, you need to enter your PIN code correctly to access your account. If a user fails more than three times when entering PIN, the machine will block the card. Assume that the user PIN is 1234, and write a Java program that asks the user to enter for the PIN no more than three times. If the user enters a wrong number three times, print a message saying "Your bank card is blocked", and end the program.

```
Please enter your PIN code: 1234
Your pin is correct.

Please enter your PIN code: 36674
Your PIN is incorrect.
Please enter your PIN code: 4321
Your PIN is incorrect.
Please enter your PIN code: 9756
Your bank card is blocked.
```

II. QUESTION (18 POINTS)

Write a program that asks the user to enter a long integer. Then sum up every digit. If the sum has two or more digits, sum up the digits again. Continue summing until obtaining a single digit. Display the result after each summation step is completed.

```
Please enter a long integer: 438857
Sum: 35
Sum: 8

Please enter a long integer: 999999999999
Sum: 108
Sum: 9
```

III. QUESTION (18 POINTS)

Write a program that gets an integer number from the user and returns the number of parity (even or odd) switches between the digits (that is, the number of pairs of consecutive digits where one is even and the other is odd, in either order). For example, if user enters 36487, the program prints 2 because the parity switches between the 3 and the 6 and again between the 8 and the 7. If the number contains only one digit, or only even or odd digits, the program prints 0. You should check that the number entered will be greater than or equal to 1.

```
Enter a number: 800
Count of Parity Switches: 0

Enter a number: 12345
Count of Parity Switches: 4
```

IV. QUESTION (16 POINTS)

Write a program that gets an integer value min and rolls a 6-sided die until it gets four consecutive rolls in a row that have values of min or less than min. As the program rolls the die it should print each value rolled, and then it should print a message at the end to indicate how many rolls were made. In the first example, notice that the program stops after rolling 3, 2, 1, and 2 consecutively because these are all values of 3 or less than 3.

```
Enter a min value: 3
524613553212
Finished after 12 rolls!

Enter a min value: 5
1363554
Finished after 7 rolls!
```

V. QUESTION (16 POINTS)

Write a program that asks user to enter an odd integer M. The program then prints (M + 1) / 2 lines where (i) each line contains M characters; (ii) N'th line has a $N-1$ '*' characters in the beginning and in the end; (iii) the remaining characters are "$+$".

```
Enter size: 5
+++++
*+++*
**+**

Enter size: 7
+++++++
*+++++*
**+++**
***+***
```

VI. QUESTION (16 POINTS)

Write a program that will generate a sequence of numbers according to the following algorithm. Your program will take an integer n. If n is even, the program will divide it by 2. If n is odd, you will multiply it by 3 and add 1. You repeat this process until you get 1.

```
Enter number: 22
22 11 34 17 52 26 13 40 20 10 5 16 8 4 2 1

Enter number: 1024
1024 512 256 128 64 32 16 8 4 2 1
```